Data Science Final Project Report

Clustering Countries Using K-Means for HELP International

Prepared by: Rafi Baihaqi
Sanbercode Data Science Program

1. Problem Introduction

1.1 Background

HELP International is an international humanitarian NGO committed to fighting poverty and providing basic facilities and assistance to people in underdeveloped countries during disasters and natural calamities.

HELP International has successfully raised approximately $10 million USD in funding. The CEO of the NGO now faces the critical challenge of strategically and effectively allocating this budget to countries that are in the most dire need of humanitarian aid.

1.2 Objective

The primary objective of this project is to categorize countries using socio-economic and health factors that determine the overall development of a nation. Specifically:

- Apply K-Means clustering to group countries based on their development indicators
- Determine the optimal number of clusters using the Elbow Method and Silhouette Score
- Identify which cluster contains the most underdeveloped countries
- Recommend countries that should be the primary focus for aid distribution

1.3 Problem Statement

Given a dataset of 167 countries with socio-economic and health indicators, the task is to use unsupervised machine learning (K-Means clustering) to segment countries and identify those with the greatest need for humanitarian assistance, specifically by focusing on per-capita income (Pendapatan) and child mortality rate (Kematian_anak).

2. Dataset

2.1 Data Source

The dataset used in this project is Data_Negara_HELP.csv, which contains socio-economic and health data for 167 countries. The dataset was provided as part of the Sanbercode Data Science Final Project.

2.2 Dataset Overview

Attribute	Value	Description
Total Records	167	Number of countries
Total Features	10	9 numerical + 1 categorical (Negara)
Missing Values	0	No missing data found in any column
Numerical dtypes	float64, int64	7 float64 columns, 2 int64 columns
Categorical dtype	object	1 object column (Negara / country name)

2.3 Descriptive Statistics

Key statistical highlights from df.describe():

- Kematian_anak: min = 2.6, max = 208.0, mean ≈ 38.3, std ≈ 40.3
- Ekspor: min = 0.109, max = 200.0, mean ≈ 41.1
- Pendapatan: min = 609, max = 125,000, mean ≈ 17,144
- Harapan_hidup: min = 32.1, max = 82.8, mean ≈ 70.6
- GDPperkapita: min = 231, max = 105,000, mean ≈ 12,964

3. Features and Processing

3.1 Feature Description

Feature	Description
Negara	Country name
Kematian_anak	Deaths of children under 5 per 1,000 live births
Ekspor	Exports of goods and services per capita
Kesehatan	Total health expenditure per capita
Impor	Imports of goods and services per capita
Pendapatan	Net income per person
Inflasi	Annual growth rate of total GDP
Harapan_hidup	Average life expectancy at birth (years)
Jumlah_fertiliti	Average number of children per woman
GDPperkapita	GDP per capita (total GDP / total population)

3.2 Missing Value Handling

A check for null/empty values was performed using df.isnull().sum(). The result confirmed that all 10 columns had zero missing values, so no imputation or row removal was required.

3.3 Multivariate Analysis – Correlation Heatmap

A correlation heatmap was generated using sns.heatmap(numeric_df.corr()) to understand the relationships between all numerical features. Key findings:

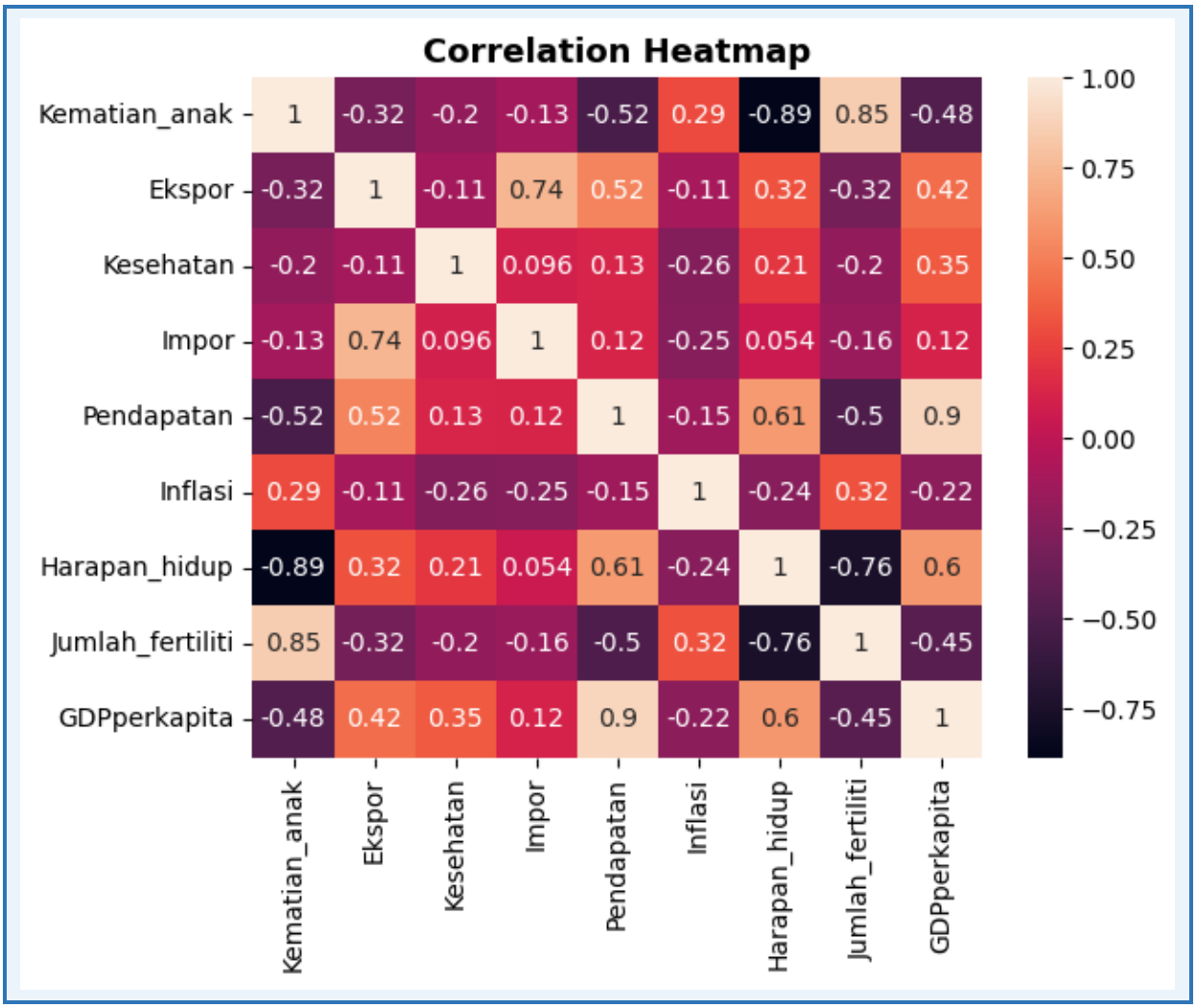
- Kematian_anak and Harapan_hidup: strong negative correlation (-0.89)
- Kematian_anak and Jumlah_fertiliti: strong positive correlation (0.85)
- Pendapatan and Kematian_anak: moderate negative correlation (-0.52)
- Pendapatan and GDPperkapita: strong positive correlation (0.90)

Based on this analysis, Pendapatan and Kematian_anak were selected as the two features for clustering. This pair was chosen because it directly represents both the socio-economic dimension (income) and the health dimension (child mortality), which aligns with the project objective. Their correlation of -0.52 is significant enough to show a meaningful relationship, while still being distinct enough to form well-separated clusters.

Although the dataset contains multiple numerical features, this study focuses specifically on Pendapatan (income per capita) and Kematian_anak (child mortality rate) for the clustering process. These two features were selected because they directly represent the two primary dimensions of the problem: economic capacity and public health conditions.

In addition, limiting the model to two features improves interpretability and visualization clarity, allowing the clustering results to be easily analyzed and communicated through 2D plots. Using all available features could increase model complexity and reduce interpretability without necessarily improving the clarity of cluster separation for this specific objective.

Therefore, this feature selection approach ensures a balance between model simplicity, interpretability, and relevance to the humanitarian aid allocation goal.



Source: sns.heatmap(numeric_df.corr(), annot=True, fmt='.2g')

3.4 Outlier Detection and Removal

Outliers in Pendapatan and Kematian_anak were detected using the IQR method via the function finding_outlier_IQR(). Values below Q1 – 1.5×IQR or above Q3 + 1.5×IQR were flagged as outliers.

Outliers detected:

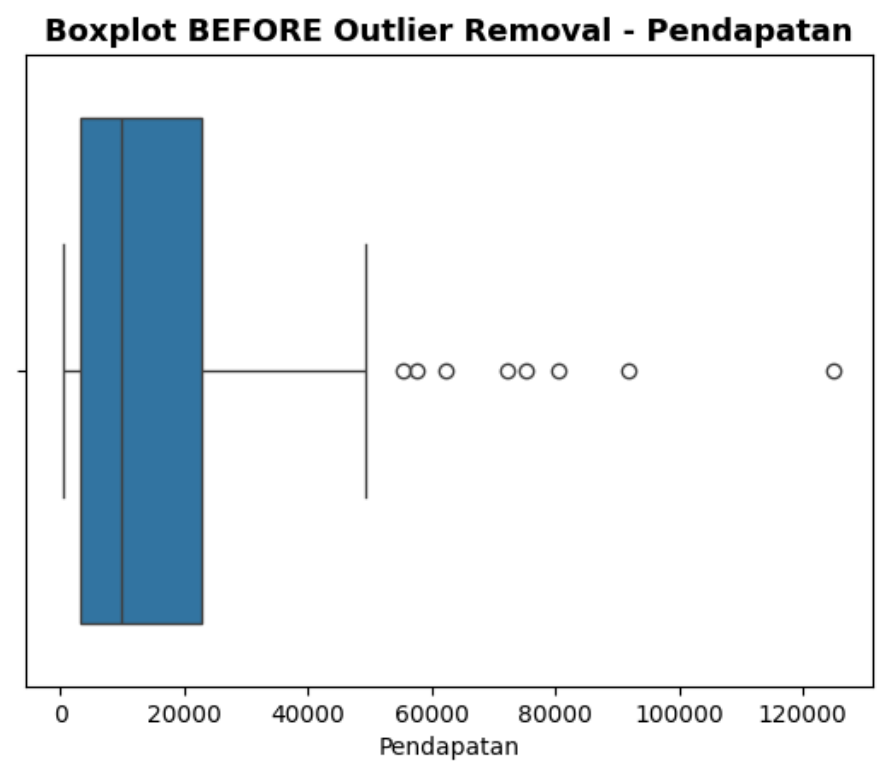
- Pendapatan: 8 outlier values (e.g., 80,600 – 125,000)
- Kematian_anak: 4 outlier values (e.g., 149, 150, 160, 208)

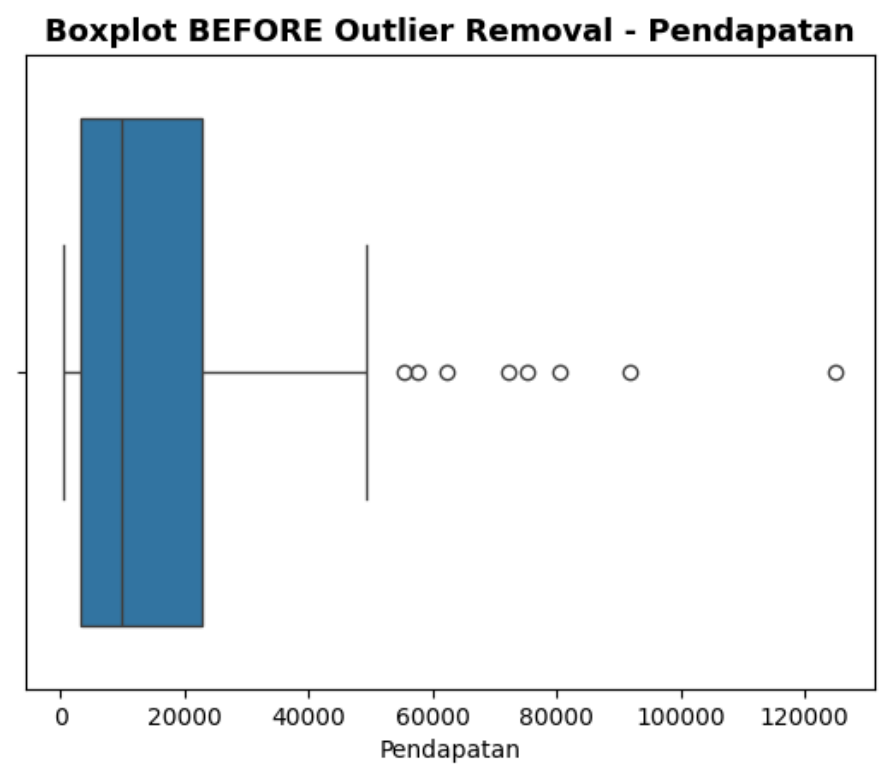
These outliers were then removed using remove_outlier_IQR(), applied to both columns simultaneously. The cleaned dataset retains rows where both Pendapatan AND Kematian_anak are within normal range, resulting in 155 rows (down from 167). The clean index was preserved to keep all original columns intact (df_clean_all_cols).

Outlier removal is an important preprocessing step for K-Means clustering because the algorithm relies on distance-based calculations (Euclidean distance). Extreme values can disproportionately influence the position of cluster centroids, leading to biased cluster formation and reducing the overall quality of segmentation.

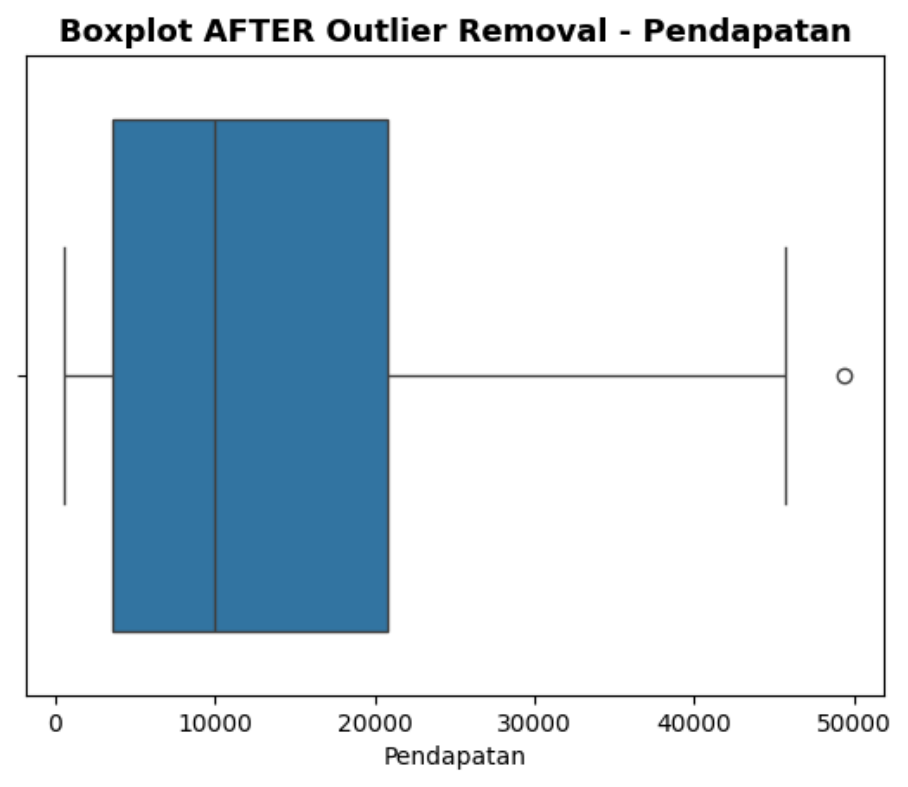
In this study, outliers in Pendapatan and Kematian_anak were identified using the Interquartile Range (IQR) method and subsequently removed. This ensures that the resulting clusters better represent the general distribution of the data rather than being skewed by a small number of extreme observations.

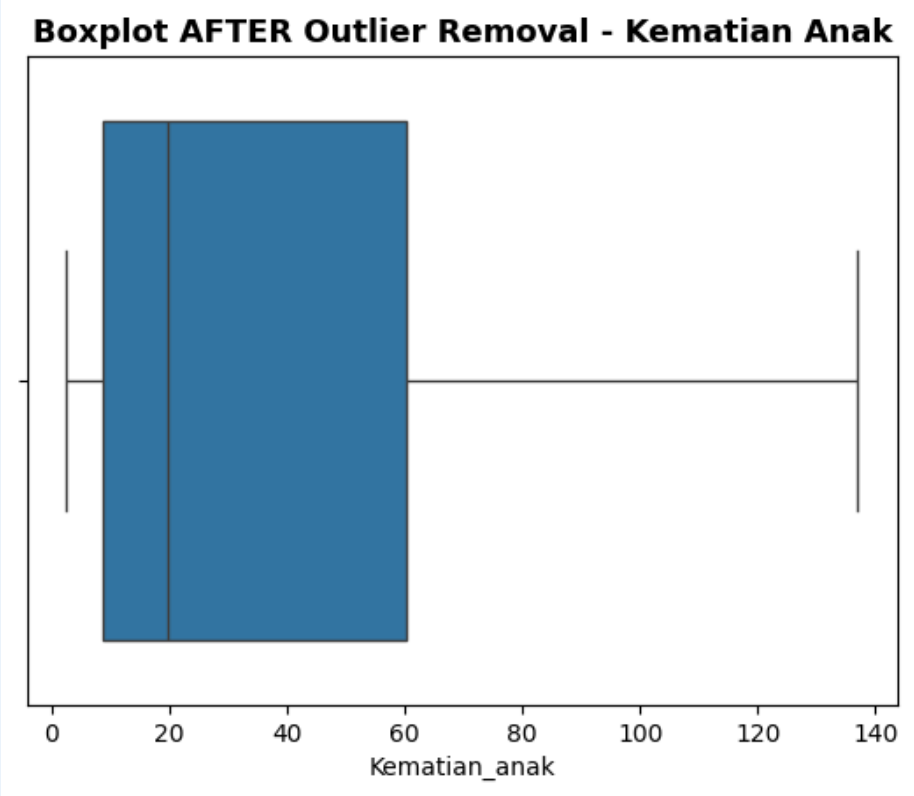
By removing outliers, the model produces more stable and representative centroids, leading to improved cluster separation and more reliable insights for decision-making.





Source: sns.boxplot(x='Pendapatan', data=df) and sns.boxplot(x='Kematian_anak', data=df). Title: ' Boxplot BEFORE Outlier Removal – Pendapatan' and Title: ' Boxplot BEFORE Outlier Removal – Kematian Anak'



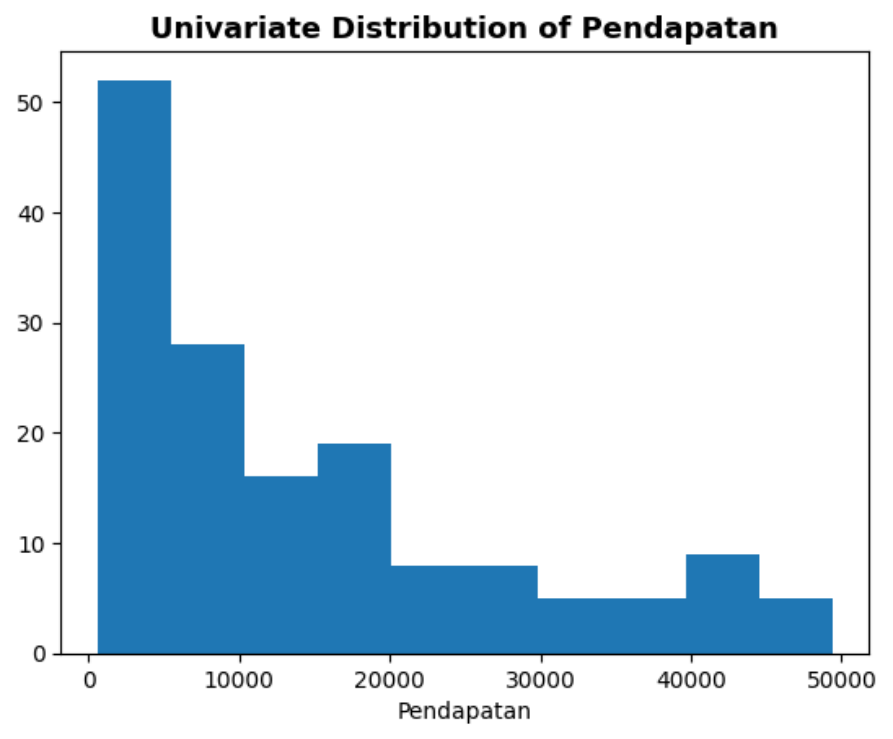


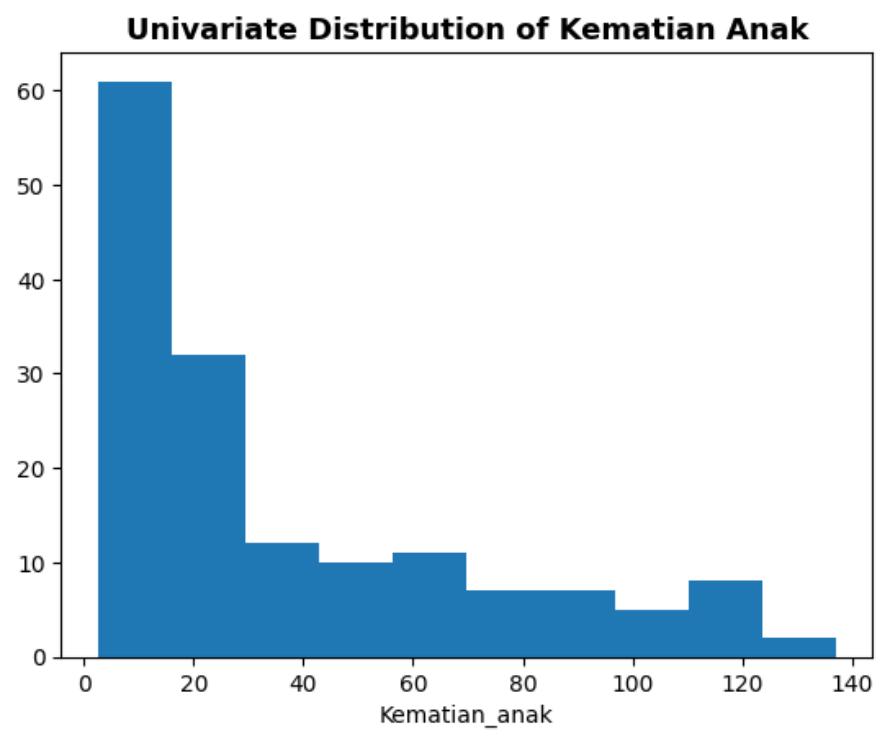
Source: sns.boxplot(x='Kematian_anak', data=df_removed_outlier) and sns.boxplot(x='Pendapatan', data=df_removed_outlier). Title: 'Visualisasi data setelah menghandling outlier'

3.5 Univariate Analysis

Histogram plots were generated for both Pendapatan and Kematian_anak after outlier removal to understand the individual distribution of each feature.

- Pendapatan: heavily right-skewed; most countries have income in the 0–10,000 range
- Kematian_anak: also right-skewed; majority of countries have child mortality rates of 0–20 per 1,000 births



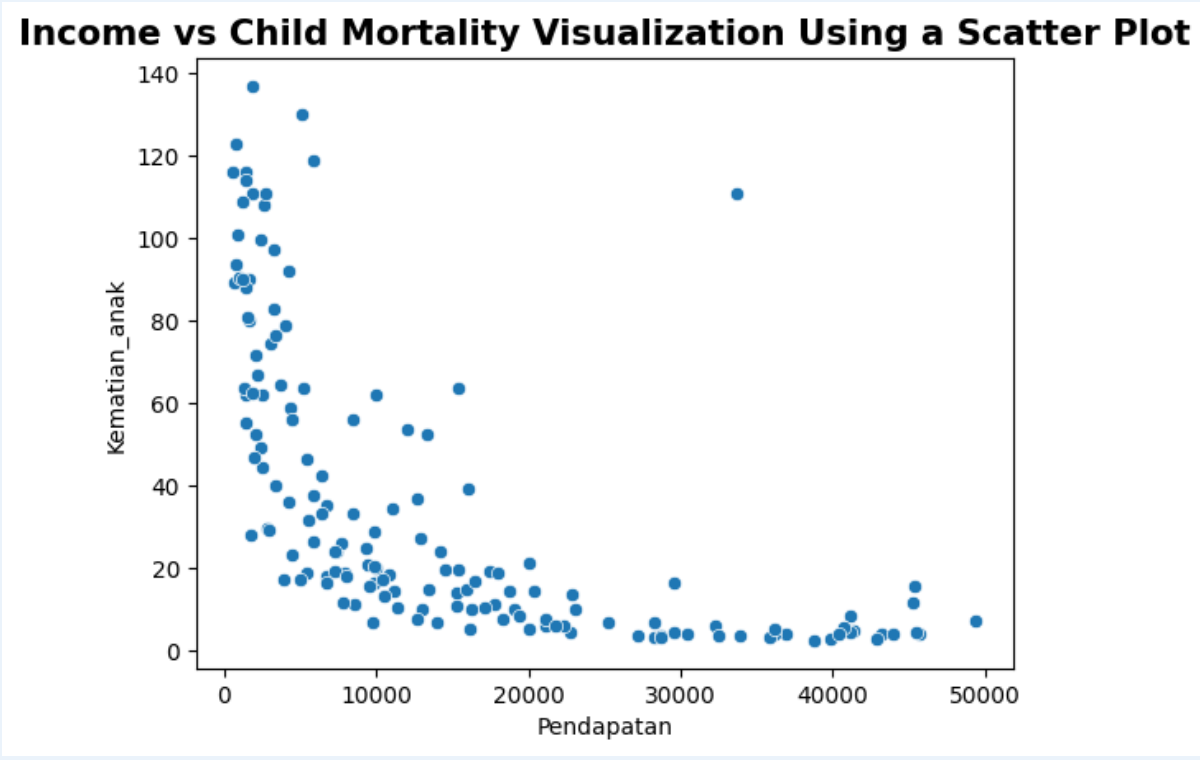


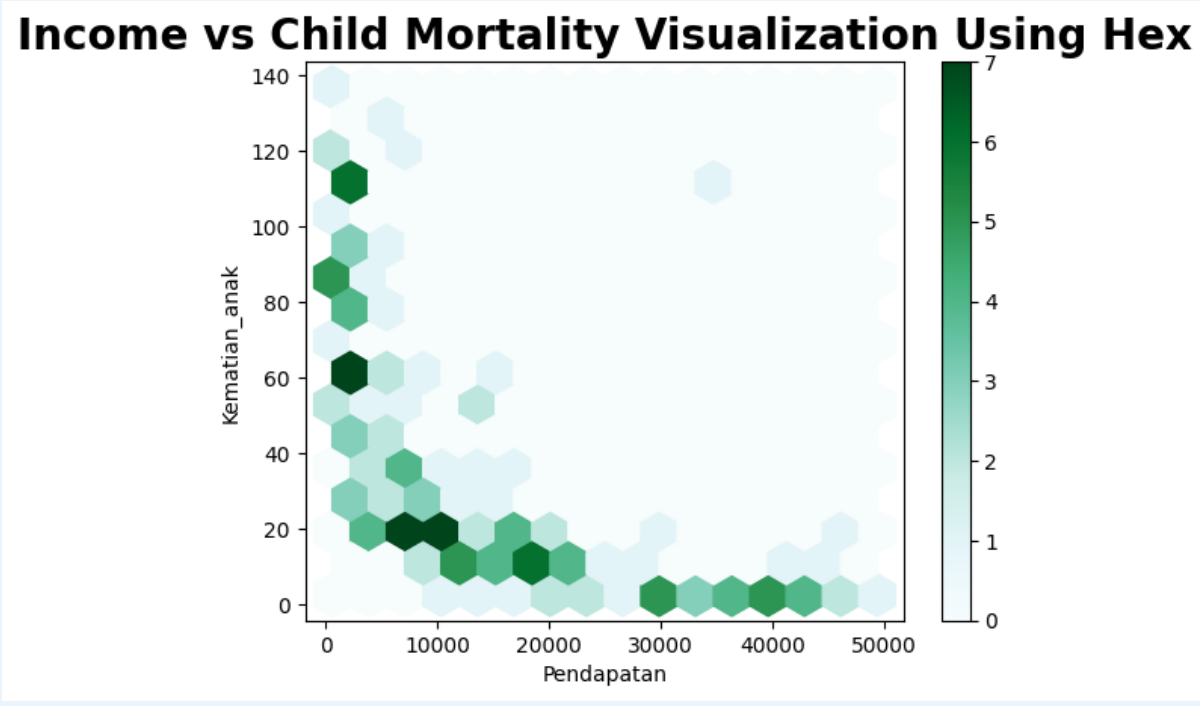
Source: ax.hist(df_removed_outlier['Pendapatan']) and ax.hist(df_removed_outlier['Kematian_anak'])

3.6 Bivariate Analysis

Two visualizations were created to examine the relationship between Pendapatan and Kematian_anak:

- Scatter plot (sns.scatterplot): confirms a clear negative trend — higher income countries tend to have lower child mortality
- Hexbin plot (df_removed_outlier.plot.hexbin, gridsize=15): shows data density, with the highest concentration of countries clustered at low income (0–10,000) and low-to-moderate child mortality (0–40)





Source: sns.scatterplot + df_removed_outlier.plot.hexbin(x='Pendapatan', y='Kematian_anak', gridsize=15)

3.7 Data Scaling

Before clustering, both features were standardized using StandardScaler (sc.fit_transform) to ensure equal contribution to the distance metric used by K-Means. The scaler was saved (sc) for later use in inverse transforming the results back to original values.

4. Models and Techniques

4.1 Algorithm: K-Means Clustering

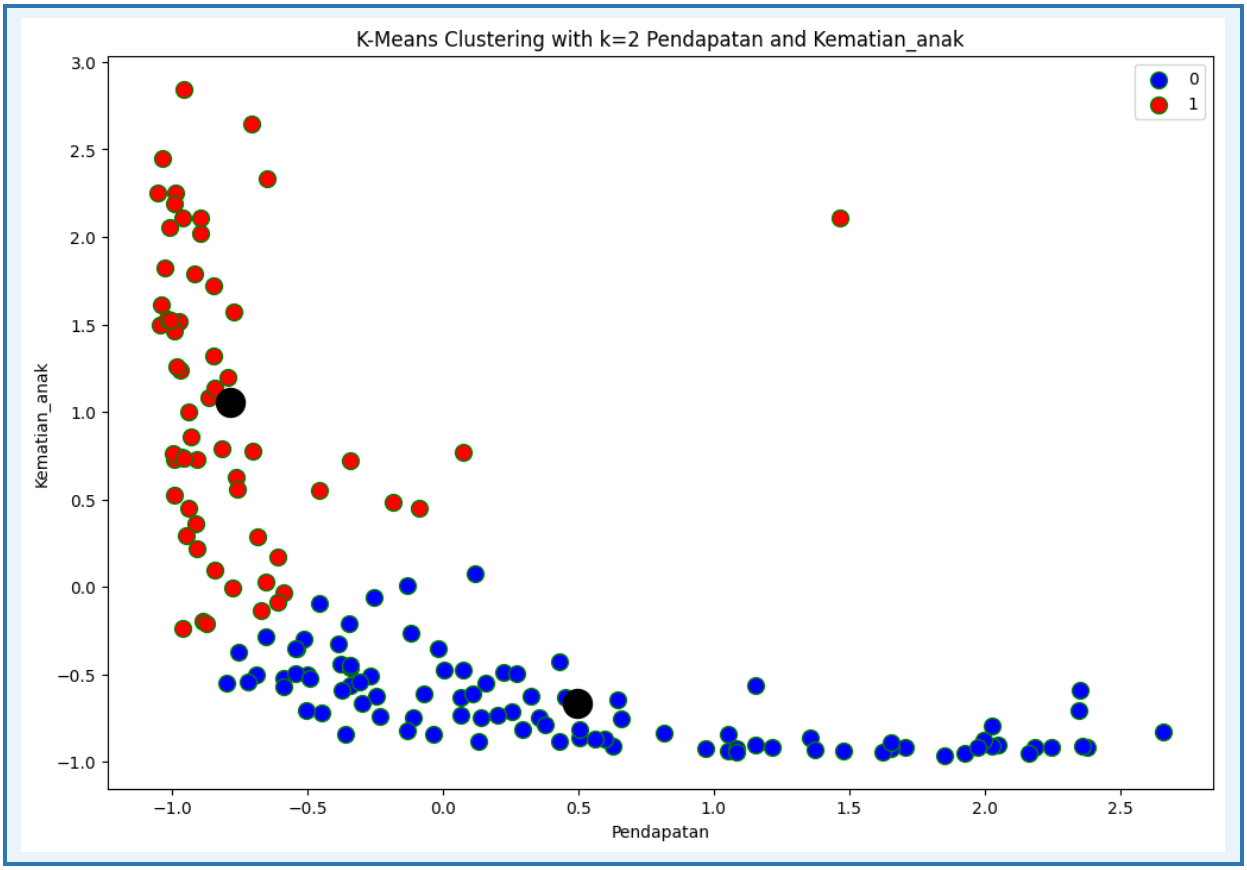
K-Means is an unsupervised machine learning algorithm that partitions data into k distinct clusters based on feature similarity. The algorithm iteratively:

1. Initializes k cluster centroids
2. Assigns each data point to the nearest centroid
3. Recalculates centroids as the mean of assigned points
4. Repeats steps 2–3 until convergence

All models used random_state=42 and n_init=10 for reproducibility and stability. The initial clustering attempt used k=2 before the optimal k was determined.

4.2 Initial Clustering: k = 2

The first clustering run used KMeans(n_clusters=2, random_state=42, n_init=10). The result produced two groups, but the Elbow Method was then applied to determine whether a different k would be more appropriate.



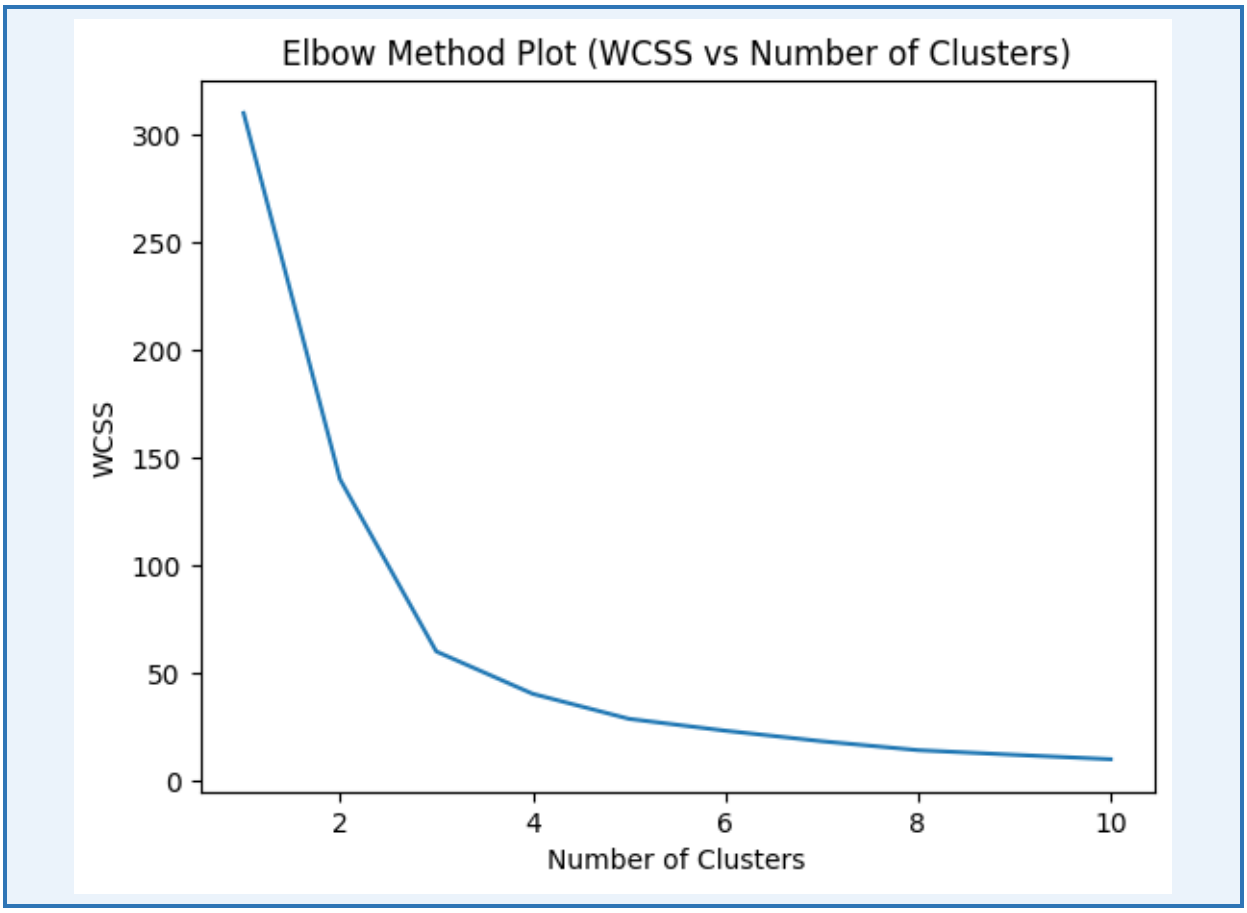
Source: scatter plot of new_dfoutlier_std, colored blue (label=0) and red (label=1), with black centroids. Title: 'Clustering Pendapatan dan Kematian_anak'

4.3 Elbow Method – Optimal k Selection

WCSS (Within-Cluster Sum of Squares) was computed for k = 1 to 10 using KMeans with init='k-means++'. The elbow in the WCSS curve indicates the optimal number of clusters where adding more clusters yields diminishing improvement.

k (Clusters)	WCSS (approx.)	Note
1	310.0	No clustering
2	140.0	Significant drop
3	60.0	★ Elbow point — optimal
4	40.3	Diminishing return
5+	< 30	Over-segmentation

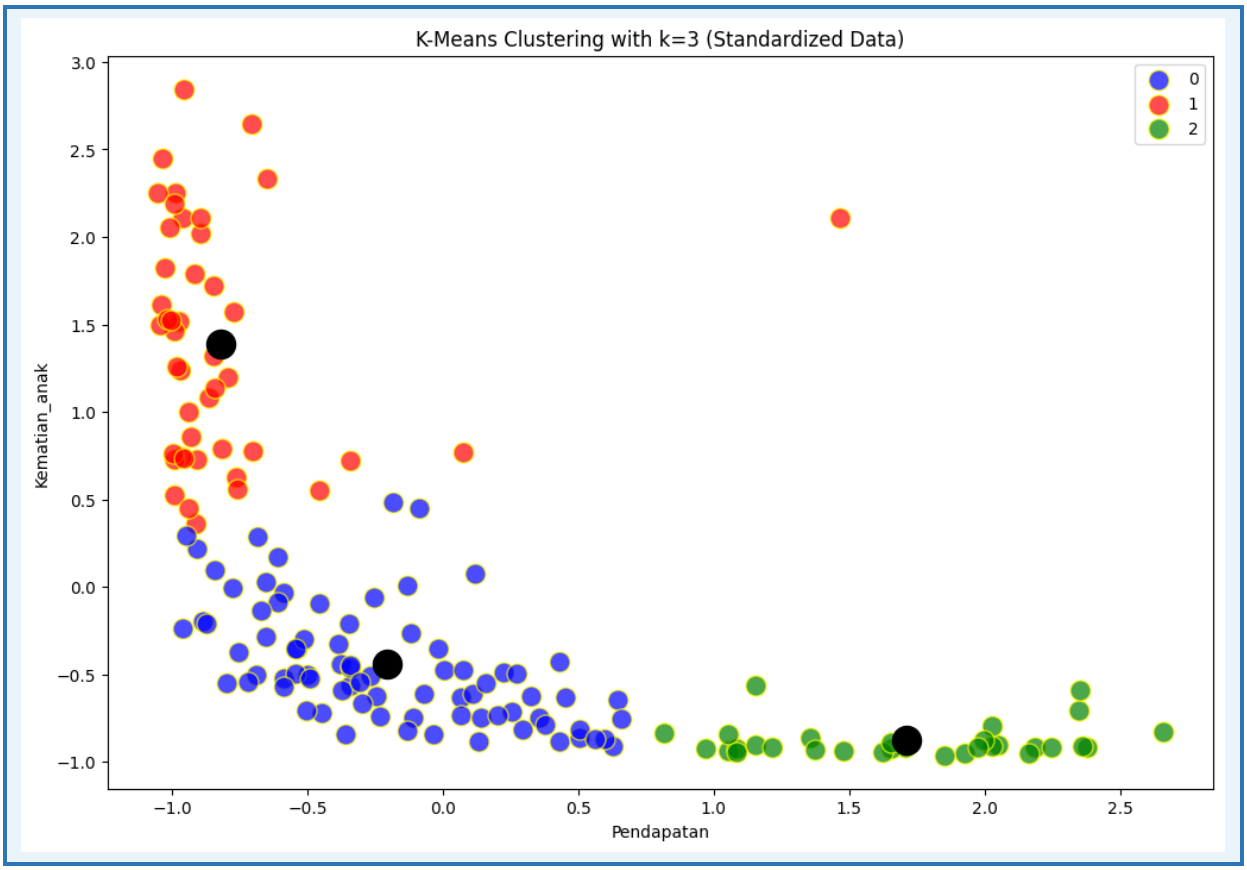
The elbow is clearly visible at k=3, indicating that 3 clusters provide the best balance between compactness and simplicity.



Source: plt.plot(range(1, 11), wcss). Title: 'Elbow Method'

4.4 Final Clustering: k = 3

The final model used KMeans(n_clusters=3, random_state=42, n_init=10) fitted on the standardized data (dfoutlier_std).



Source: Scatter plot of new_dfoutlier_std colored blue (0), red (1), green (2), with black centroids. Title: 'K-Means Clustering with k=3 (Original / Actual Values)'

4.5 Model Validation: Silhouette Score

The Silhouette Score measures how similar each data point is to its own cluster compared to other clusters. Values range from -1 (wrong cluster) to +1 (perfect separation), with higher being better.

k	Silhouette Score	Interpretation
2	0.4861	Moderate cluster separation
3	0.5569	★ Better defined clusters — preferred

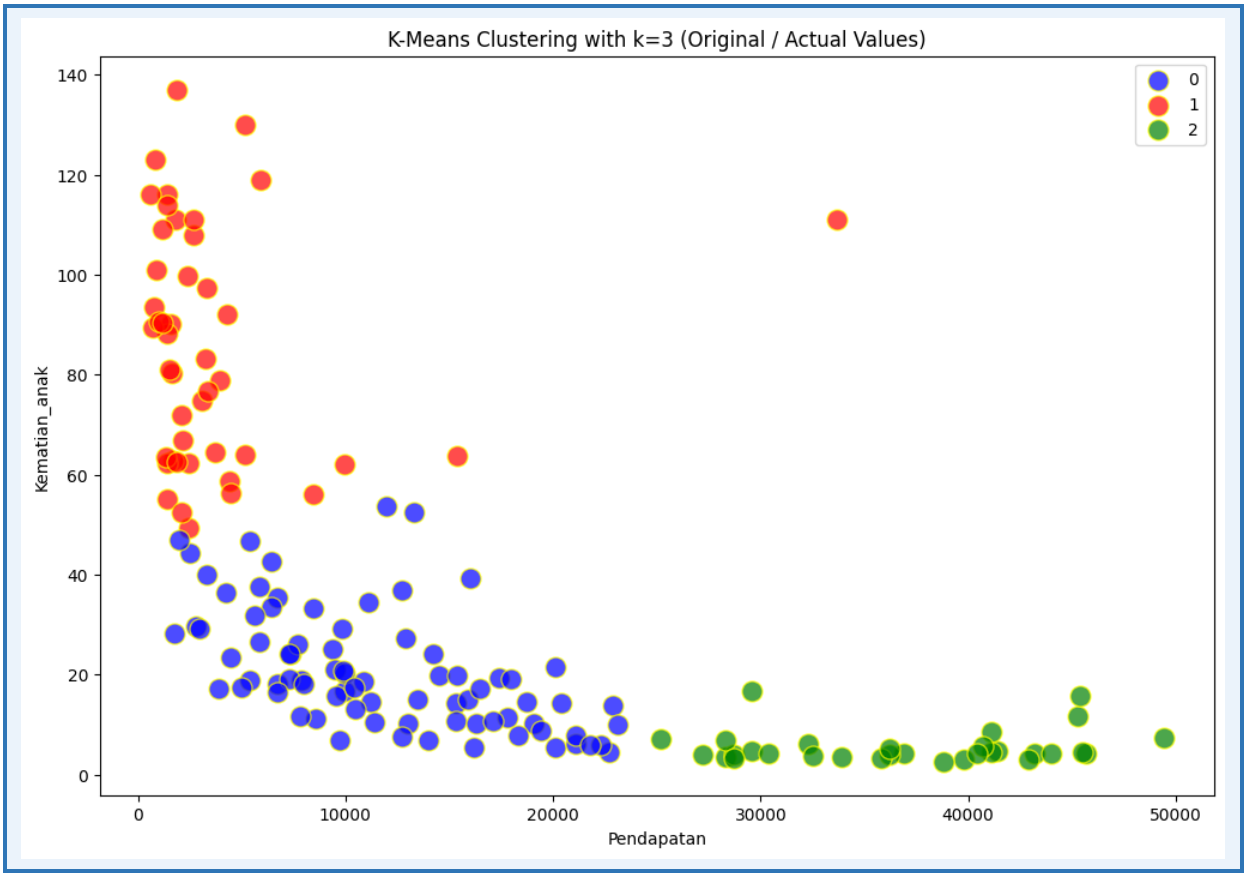
The higher Silhouette Score at k=3 (0.5569 > 0.4861) confirms that 3 clusters provide clearer and more meaningful segmentation than 2. This, combined with the Elbow Method, validates k=3 as the optimal choice.

5. Results and Discussion

5.1 Cluster Interpretation

After applying K-Means with k=3, the results were inverse-transformed using sc.inverse_transform() to plot and analyze in original units. The clusters were then attached to the full dataset (df_clean_all_cols) via the label column.

Cluster	Income Level	Child Mortality	Interpretation
0 (Blue)	Medium – High	Low	Developing to developed countries
1 (Red)	Low	High	Underdeveloped — PRIORITY for aid
2 (Green)	High	Very Low	Well-developed countries



Source: sc.inverse_transform on dfoutlier_std, plotted as newdf. Red (1) drawn first to avoid overlap. Title: 'Clustering Pendapatan dan Kematian_anak'

5.2 Dataset Reference Statistics

The following mean values from the full cleaned dataset (df_clean_all_cols) were computed as reference for interpreting cluster boundaries:

- Mean Kematian_anak: 38.84 deaths per 1,000 births
- Mean Pendapatan: 17,145.56 per capita

Countries in Cluster 1 (Red) fall well below the mean income and well above the mean child mortality, confirming this cluster as the underdeveloped group.

5.3 Recommended Countries for Aid

Countries are filtered from Cluster 1 only (cari_negara = df_clean_all_cols[df_clean_all_cols['label']==1]). A combined priority score was computed using MinMaxScaler normalization on both features: Score = (1 – Pendapatan_norm) + Kematian_norm, ensuring both dimensions contribute equally regardless of scale. The 8 countries with the highest combined scores are:

No.	Country	Child Mortality	Income (Pendapatan)	Priority Basis
1	Sierra Leone	160.0	1,220	Highest child mortality
2	Congo, Dem. Rep.	116.0	609	Lowest income AND highest child mortality
3	Burkina Faso	116.0	1,430	Highest child mortality
4	Angola	119.0	5,900	Highest child mortality
5	Benin	111.0	1,820	Highest child mortality
6	Mozambique	101.0	918	Lowest income
7	Lesotho	99.7	2,380	High child mortality
8	Liberia	89.3	700	Lowest income

5.4 Discussion

The scatter plot of the original (inverse-transformed) values confirms a clear negative relationship between income (Pendapatan) and child mortality (Kematian_anak). Countries in Cluster 1 are concentrated in the lower-income and higher-mortality region, making them both visually and analytically distinct from the other clusters.

The list of recommended countries is dominated by nations in Sub-Saharan Africa, such as Sierra Leone, Congo (Dem. Rep.), Burkina Faso, Angola, and Mozambique. This pattern is consistent with global development trends, where countries in this region often face significant challenges in both economic capacity and healthcare outcomes.

The prioritization is based on a combined score derived from normalized income and child mortality, allowing both dimensions to contribute equally. This approach highlights countries that simultaneously experience low income and high child mortality, rather than focusing on a single extreme indicator.

Sierra Leone emerges as the highest priority country due to its extremely high child mortality rate combined with low income. Congo (Dem. Rep.) also stands out as it combines the lowest income with very high mortality, reinforcing its urgency for aid intervention.

Overall, this approach provides a balanced and data-driven prioritization by integrating both socio-economic and health indicators into a single composite score.

While Cluster 1 is identified as the primary target for humanitarian aid due to its combination of low income and high child mortality, it is equally important to understand why the other clusters are not prioritized.

Countries in Cluster 2 (high income, very low child mortality) already demonstrate strong economic capacity and well-established healthcare systems. Allocating significant aid to this group would result in inefficient resource utilization, as these countries are relatively capable of addressing their own development challenges.

Meanwhile, Cluster 0 (medium to high income, low child mortality) represents countries in a transitional or developing stage. Although they may still face certain challenges, their overall conditions are considerably more stable compared to Cluster 1. As a result, they are not considered urgent priorities for immediate humanitarian intervention.

A misallocation of resources—such as directing funds toward Cluster 0 or Cluster 2—could lead to opportunity loss, where countries in critical condition (Cluster 1) receive insufficient support. This may result in preventable outcomes, including higher child mortality rates and slower socio-economic recovery in the most vulnerable regions.

Therefore, focusing aid distribution on Cluster 1 ensures that resources are directed toward countries with the highest urgency and greatest potential impact, maximizing the effectiveness of the available budget.

6. Conclusions

1. The correlation heatmap identified Pendapatan and Kematian_anak as the most relevant feature pair for clustering, with a correlation of -0.52 representing the socio-economic and health dimensions of development.
2. Outlier removal via IQR reduced the dataset from 167 to 155 countries, preserving data quality without losing geographic diversity.
3. The Elbow Method identified k=3 as the optimal cluster count, confirmed by a Silhouette Score of 0.5569 for k=3 versus 0.4861 for k=2.
4. Three clusters emerged: underdeveloped (low income, high mortality), developing/transitional (medium income), and well-developed (high income, low mortality).
5. Eight countries from Cluster 1 are recommended as priority aid recipients, with Sierra Leone identified as the top priority due to its extremely high child mortality (160.0) and low income (1,220). Congo (Dem. Rep.) also represents a critical case, combining the lowest income (609) with very high child mortality (116.0), making it one of the most urgent targets for humanitarian assistance.

6.1 Strategic Recommendations for CEO

- Immediate health intervention focus: Sierra Leone, Angola, and Burkina Faso (very high child mortality rates above 110).
- Economic development focus: Congo (Dem. Rep.) and Mozambique (extremely low income levels).
- Balanced health + economic support: Benin, Liberia, and Lesotho (moderate combination of both risk factors).
- Future work: incorporate additional features (education index, access to clean water, political stability) for richer clustering and more comprehensive prioritization.

6.2 Limitations

- Only two features used for clustering; a more comprehensive model would leverage all 9 numerical features.
- Outlier removal dropped 12 records; some extreme cases (e.g., very high child mortality) may have been excluded.
- The dataset is a static snapshot; development indicators change over time and require periodic re-analysis.